

Kubernetes - Complements and evaluation

Install Helm 3

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam$ curl https://raw.githubusercontent.com/helm/helm/master/scripts/get-helm-3 | bash
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 11913 100 11913 0 0 121k 0 --:--:-- --:--:-- --:--:-- 121k
Downloading https://get.helm.sh/helm-v3.18.3-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
helm installed into /usr/local/bin/helm
```

Add official chart repo

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam$ helm repo add stable https://charts.helm.sh/stable
"stable" has been added to your repositories
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "stable" chart repository
Update Complete. *Happy Helming!*
```

Bootstrap own chart

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam$ helm create datascientest-chart
Creating datascientest-chart
```

Prune out the unused boilerplate

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam$ cd datascientest-chart/templates
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart/templates$ rm _helpers.tpl serviceaccount.yaml NOTES.txt
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart/templates$ rm -rf tests
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart/templates$ ls
deployment.yaml hpa.yaml ingress.yaml service.yaml
```

Create files in datascientest-chart/

- *dev-values.yaml*
- *prod-values.yaml*
- *staging-values.yaml*
- *values.yaml*

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ ls
Chart.yaml charts dev-values.yaml prod-values.yaml staging-values.yaml templates values.yaml
```

Create namespaces

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl create namespace dev
namespace/dev created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl create namespace staging
namespace/staging created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl create namespace prod
namespace/prod created
```

Update code in file exam/datascientest-chart/templates/_helpers.tpl

Ensure \$KUBECONFIG is set

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ export KUBECONFIG=$HOME/.kube/config
```

Create file exam/datascientest-chart/templates/secret.yaml

Update code in files

- exam/datascientest-chart/templates/deployment.yaml
- exam/datascientest-chart/templates/service.yaml
- exam/datascientest-chart/templates/ingress.yaml
- exam/datascientest-chart/templates/hpa.yaml

Create separate directory `app/` for application sources and Dockerfile (not under Helm chart)

- `exam/app/Dockerfile`
- `exam/app/main.py`
- `exam/app/requirements.txt`

Build and push your FastAPI image

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/app$ docker build -t localhost:5000/fastapi-app:1.0.0 .
[+] Building 94.2s (9/9) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default
=> => transferring dockerfile: 372B                                              0.0s
=> [internal] load metadata for docker.io/library/ubuntu:20.04                  0.0s
=> [auth] library/ubuntu:pull token for registry-1.docker.io                    0.7s
=> [internal] load .dockerignore                                                  0.0s
=> => transferring context: 2B                                                    0.0s
=> [internal] load build context                                                  0.0s
=> => transferring context: 1.48kB                                                0.0s
=> [1/3] FROM docker.io/library/ubuntu:20.04@sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafa15388214 3.5s
=> => resolve docker.io/library/ubuntu:20.04@sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafa15388214 0.0s
=> => sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafa15388214 6.69kB / 6.69kB
=> => sha256:c664f8f86ed5a386b0a340d981b8f81714e21a8b9c73f658c4bea56aa179d54a 424B / 424B
=> => sha256:b7bab04fd9aa0c771e5720bf0cc7cbf993fd6946645983d9096126e5af45d713 2.30kB / 2.30kB
=> => sha256:13b7e930469f6d3575a320709035c6acf6f5485a76abcf03d1b92a64c09c2476 27.51MB / 27.51MB
=> => extracting sha256:13b7e930469f6d3575a320709035c6acf6f5485a76abcf03d1b92a64c09c2476 2.6s
=> [2/3] ADD requirements.txt main.py ./                                         0.4s
=> [3/3] RUN apt-get update && DEBIAN_FRONTEND=noninteractive apt-get install -y python3-pip libmysqlclient-dev & 83.7s
=> => exporting to image                                                            5.6s
=> => exporting layers                                                            5.6s
=> => writing image sha256:437f0a899498ab6a836955ff915dbddc1666b733ced26de2543bf64e500976d1 0.0s
=> => naming to docker.io, localhost:5000/fastapi-app:1.0.0                     0.0s
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/app$ docker push localhost:5000/fastapi-app:1.0.0
The push refers to repository [docker.io/localhost:5000/fastapi-app]
52137969a6ec: Pushed
e128f79ed5de: Pushed
470b66ea5123: Mounted from library/ubuntu
1.0.0: digest: sha256:b79316580df7f6e72e669c1f559f105f2f95981a0e3d7af0f3cb1fa7605594e5 size: 949
```

Ensure context and namespaces still exist

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl create ns dev || true
Error from server (AlreadyExists): namespaces "dev" already exists
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl create ns staging || true
Error from server (AlreadyExists): namespaces "staging" already exists
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl create ns prod || true
Error from server (AlreadyExists): namespaces "prod" already exists
```

Three helm upgrade `--install` each deployed (or updated) the `datascientest-chart` release—named `user-api-dev`, `user-api-stg` and `user-api-prod`—into the `dev`, `staging` and `prod` namespaces, applying config overrides in `dev-values.yaml`, `staging-values.yaml` and `prod-values.yaml`

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ helm upgrade --install user-api-dev . \
> --namespace dev \
> --values dev-values.yaml
Release "user-api-dev" has been upgraded. Happy Helming!
NAME: user-api-dev
LAST DEPLOYED: Tue Jul 8 19:47:22 2025
NAMESPACE: dev
STATUS: deployed
REVISION: 3
TEST SUITE: None
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ helm upgrade --install user-api-stg . \
> --namespace staging \
> --values staging-values.yaml
Release "user-api-stg" has been upgraded. Happy Helming!
NAME: user-api-stg
LAST DEPLOYED: Tue Jul 8 19:47:36 2025
NAMESPACE: staging
STATUS: deployed
REVISION: 3
TEST SUITE: None
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ helm upgrade --install user-api-prod . \
> --namespace prod \
> --values prod-values.yaml
Release "user-api-prod" has been upgraded. Happy Helming!
NAME: user-api-prod
LAST DEPLOYED: Tue Jul 8 19:47:50 2025
NAMESPACE: prod
STATUS: deployed
REVISION: 3
TEST SUITE: None
```

Snapshot the raw YAML

Four kubectl get ... -o yaml > *.yaml commands extracted live Secret, Deployment, Service and Ingress objects for user-api-dev-datascientest-chart release in dev namespace and saved them into local files (my-secret-eval.yaml, my-deployment-eval.yaml, my-service-eval.yaml and my-ingress-eval.yaml)

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl get secret user-api-dev-datascien
test-chart-db \
> -n dev -o yaml > my-secret-eval.yaml
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl get deploy user-api-dev-datascien
test-chart \
> -n dev -o yaml > my-deployment-eval.yaml
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl get svc user-api-dev-datascientes
t-chart \
> -n dev -o yaml > my-service-eval.yaml
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/exam/datascientest-chart$ kubectl get ingress user-api-dev-datascie
ntest-chart \
> -n dev -o yaml > my-ingress-eval.yaml
```